

Shortest Path Planning For Tennis Ball Detection

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Abstract—During tennis training sessions, coaches usually hit players with hundreds of balls to practice. At the end of the session, players often need to stop training several minutes early to manually collect the scattered tennis balls, which reduces effective training time and interrupts the practice flow. To address this problem, this project proposes an autonomous robotic system designed to detect and collect tennis balls efficiently. The system uses an Intel RealSense D415 depth camera, which provides both RGB images and depth information to understand the 3D environment. A YOLO (You Only Look Once) object detection model is applied to identify tennis balls in real time and determine their locations within the scene. Using the depth data from the camera, the system estimates the spatial coordinates of each detected ball. These coordinates are then sent to the robot, which computes an efficient path and autonomously navigates the area to collect the balls. The proposed solution aims to automate the ball collection process, reduce interruptions during training sessions, and increase overall practice efficiency through the integration of computer vision, depth sensing, and robotic path planning.

Index Terms—YOLO, Shortest Path, Object Detection, Autonomous, Depth Camera

I. INTRODUCTION

Efficient training is essential for athletes to improve their performance, but many practical challenges can limit practice time. In sports like tennis, one common challenge is managing the large number of balls scattered across the court. Manually collecting these balls takes significant time, reduces actual practice time, and interrupts the flow of the session, especially on large courts or during intensive training.

To address this challenge, this project uses deep-learning based algorithm such as YOLO model to detect balls quickly and accurately on the court. Once the ball positions are identified, path planning algorithms are applied to guide a robot in collecting the balls efficiently without human intervention.

As part of the project, the ROS 2 RViz2 simulation environment is built to test the system on a computer before real-world deployment. This allows us to refine the algorithms and ensure that the robot collects balls smoothly and safely.

This approach is not limited to tennis; it can also be applied to other sports, such as hockey, or any activity that involves gathering scattered objects. In this way, it can save significant time and effort for players and coaches.

II. LITERATURE REVIEW

Autonomous tennis ball retrieval robots have attracted more attention these days as a practical application of computer vision and robotics. These systems aim to automatically detect and collect tennis balls scattered over a court, and reduce manual effort during training sessions. Most of tennis ball collection systems are consist of two main components, a vision system for ball detection and a pah planning method that guides the robot to the ball locations.

One of the early works in this area was presented by [1], who developed an autonomous tennis ball retrieval robot designed as a learning tool for engineering students. Their system used a single RGB camera mounted at the front of the robot to capture images of the court. For ball detection, the system relied on traditional computer vision algorithms. First, the captured images were converted from RGB to HSV color space, and color thresholding was applied to isolate the yellow-green color of tennis balls. After segmentation, contour detection and circular shape filtering were used to identify objects that match the shape of tennis balls. Once the ball position was detected in the image, the robot used simple target-following algorithm to move toward the detected location and collect the ball. These methods were computationally efficient and easy to implement, but they were sensitive to lighting conditions and background colors.

Later work by Chen [2] introduced improvements in robot navigation and path planning. In this work, the robot used a forward-facing monocular camera mounted on the robot chassis to detect tennis balls. Instead of moving toward the closest ball, the system used swarm-based optimization to determine an efficient path for collecting all balls. This approach improved the efficiency of the ball collection process. Despite these improvements, the system still depended on traditional visual detection techniques such as color segmentation, which limited the detection accuracy in a complex environment. Also, variations in lighting and occlusions still affected the system's ability to detect tennis balls.

More recent research by Latifinavid and Azizi [3] proposed deep learning-based object detection methods and more intelligent control strategies for tennis ball collection robots. In this system, The robot used LiDAR and a monocular camera mounted on the robot platform, positioned to observe the

ground area in front of the robot. YOLOv5 have been used to improve the accuracy of detection the small tennis balls. Then, a fuzzy logic controller was used to guide the robot toward the detected balls and handle navigation decisions. The fuzzy logic approach allowed the robot to make smoother decisions when targeting objects or avoiding obstacles.

Another recent research by Zegang Wang [4] deppened on deep learning-based object detection methods. In his system, the robot used a vision sensor mounted on the robot to scan the environment. The detection and tracking of tennis balls were performed using a twin neural network-based tracking algorithm. This algorithm allows the robot to track objects in real time by comparing features extracted from the target object and the current frame. While for path planning, the system used an optimization-based planning algorithm Bidirectional Rapidly-exploring Random Tree (RRT) to determine the order of the balls to be collected. The system integrates target tracking with path planning to improve navigation efficiency.

Sarifudin et al. [5] presented an IoT-integrated autonomous tennis ball collecting robot using a dual-camera perception system. The system employed HSV color segmentation and contour-based detection to identify tennis balls at both long (5–15 m) and short (0.5–5 m) ranges. After detecting the balls, the robot used a coordinate-based navigation method to move toward the detected targets. The detected ball positions were converted into robot movement commands, allowing the robot to collect the balls sequentially.

III. METHODOLOGY AND IMPLEMENTATION

The overall architecture of the proposed system is shown in Figure 1. The methodology consists of seven main stages: data collection, data preprocessing, tennis ball detection using a YOLO-based deep learning model training, evaluation, Real-time deployment and 3D position estimation using depth data, simulation environment and path planning for efficient ball collection. Each stage of the system is described in detail in the following subsections.

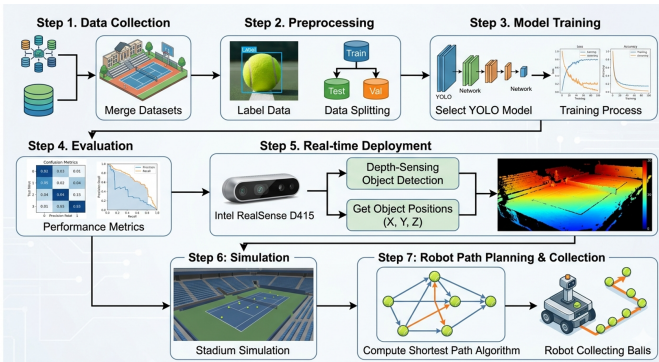


Fig. 1: The proposed system methodology Flowchart

A. Data Collection

To train and evaluate the proposed tennis ball detection model, a dataset was created by combining multiple sources.

Three datasets [6] [7] [8] were obtained from the Roboflow platform, which provides annotated datasets for computer vision tasks. In addition to these datasets, a custom dataset was collected from a real tennis court using an Intel RealSense D415 depth camera. The purpose of collecting custom data was to capture realistic environmental conditions such as different lighting situations, ball positions, multiple tennis balls are located close to each other, and court backgrounds. The total number of images obtained from all sources is summarized below.

B. Data Preprocessing

After merging the datasets, all images from Roboflow were reviewed and prepared for training. While the custom dataset was manually labeled using python code. ((Add more details about the code ???)) Figure 2 shows an example of one data point before and after labeling that used for training step.



(a) Labeling input



(b) Labeling output

Fig. 2: Example of (a) the input and (b) the output of the labeling process

The final dataset consisted of images containing tennis balls in different positions, scales, and backgrounds to improve the robustness of the detection model. Data augmentation

techniques were also applied to increase dataset diversity and improve the generalization capability of the model. The dataset was divided into three groups: Training, Validation and Testing.

C. Model Training

Multiple versions of the YOLO object detection model were evaluated for this task. After extensive experimentation, YOLOv26 [9] consistently provided the best results. YOLOv26 is an advanced real-time object detection model designed to achieve a balance between high accuracy and fast inference speed. It builds upon previous YOLO architectures by incorporating a more efficient backbone for feature extraction, a modified neck for multi-scale feature aggregation, and improved detection heads that enhance localization and classification performance as visually summarized in Figure 3. The backbone uses a combination of convolutional layers and residual connections to capture both low-level and high-level features, while the neck combines features from different stages to better detect objects of varying sizes. The detection heads then predict bounding boxes, objectness scores, and class probabilities across multiple scales, allowing the model to robustly identify small and densely packed objects. Within YOLOv26, we tested different variants including Nano, Small, and Medium, and found that they all performed well, with YOLOv26 achieving the highest accuracy, particularly in scenarios with multiple balls located close together on the court. For model training, we utilized a workstation equipped with

dataset collected, including scenarios with clustered tennis balls.

D. Evaluation

((add the evaluation metrics that used ???))

E. Real-time deployment

F. Simulation environment

G. Path Planning

IV. RESULTS

V. CONCLUSION

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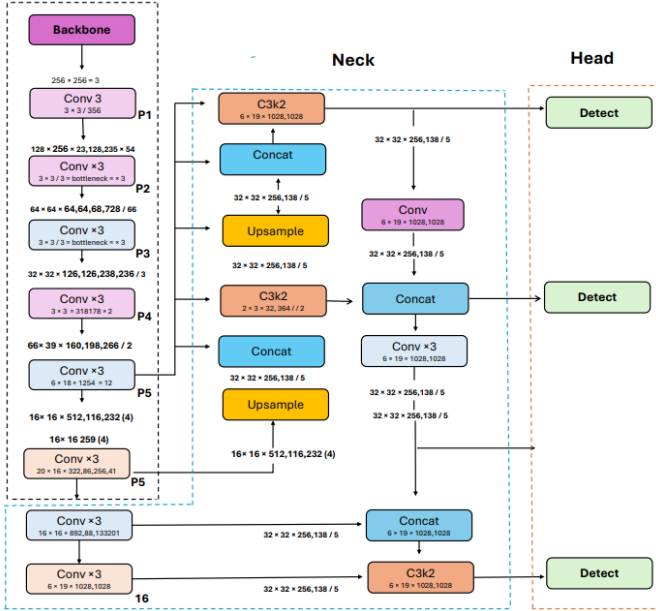


Fig. 3: Core architecture diagram of YOLOv26 object detection and segmentation algorithm

the following hardware: an NVIDIA RTX 4090 GPU, 64 GB of RAM, and an Intel Core i9 CPU. The GPU acceleration enabled faster training and efficient handling of the large